

EDUCATION

- **University of Wisconsin, Madison** Madison, WI
PhD in Electrical and Computer Engineering
◦ **Research:** High-Performance Computing, Graph Algorithms
Sept. 2023 – Present
- **University of Utah** Salt Lake City, UT
Master of Entertainment Arts and Engineering
Aug. 2020 – May. 2022
- **National Cheng Kung University** Tainan, Taiwan
Bachelor of Science in Computer Science
Sept. 2015 – June. 2019

EXPERIENCE

- **Student Researcher** Google, Mountain View, CA
AI² Chips Team
May 2024 – Aug. 2024
 - Designed algorithms that parallelize SPICE simulation while pruning redundant input patterns to improve in-house timer performance.
 - Achieved up to **20× speedup** when analyzing timing on various designs.
- **Software Engineer Intern** Intel Corporation, Santa Clara, CA
Design Methodology and Automation Team
May 2022 – Aug. 2022
 - Utilized a constraint analyzer to automate FPGA design debugging.
- **Software Testing Intern** Collins Aerospace, Salt lake City, UT
Software Verification Team
Feb. 2021 – Dec. 2021
 - Built a regression test tracking system to automatically generate test plans for QA engineers.

SELECTED PUBLICATIONS

- **G-PathGen: An Efficient GPU-parallel Critical Path Generation Algorithm (ICS'26)**
 - Designed efficient GPU kernel algorithms to accelerate parallel critical path generation.
 - Dynamically adjusted the generated path count to maximize GPU utilization while minimizing redundant work.
 - Achieved **1.6–243.8× speedup** compared to a state-of-the-art GPU solution and **100% accuracy** when generating one million paths on large designs.
- **PathGen: An Efficient Parallel Critical Path Generation Algorithm (ASP-DAC'25)**
 - Built a CPU-parallel algorithm that groups paths with similar slack so multiple threads can explore them simultaneously.
 - Designed a re-balancing algorithm to keep each group balanced, reducing thread contention and maximizing memory speed when they access the same group.
 - Achieved **2.7–7.4× speedup** with 16 threads and **nearly 100% accuracy** compared to a state-of-the-art single-threaded solution when generating one million paths on large designs.
 - **Best Paper Nominee (13 out of 169 accepted papers)**
- **Ink: Efficient Incremental k-Critical Path Generation (DAC'24)**
 - Designed an efficient algorithm that identifies a set of paths to reuse for the next query and effectively prunes the path search space.
 - Achieved **5.2–22.4× speedup** and **20–31% less memory usage** compared to a state-of-the-art timer when generating one million paths on large designs.
- **G-PASTA: GPU-Accelerated Partitioning Algorithm for Static Timing Analysis (DAC'24)**
 - Built a GPU-accelerated algorithm that splits large timing dependency graphs into smaller subgraphs to reduce processor scheduling cost.
 - Achieved up to **41.8× faster partitioning** and improved total static timing analysis runtime by **43%** on large designs.

PROGRAMMING SKILLS

- C++, CUDA